Reg. No. | | | | | | | | | | |

[17ME42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

BASIC THERMODYNAMICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least TWO questions from each part.
2. Missing data may be suitably assumed.
3. Use of thermodynamics data hand book, steam tables and Mollier chart are permitted.

PART - A

1. a. Distinguish between: **04**
   i. Intensive and extensive properties
   ii. Thermal and thermodynamic equilibrium

   b. Classify the below examples into open system, closed system, and isolated Systems. **03**
   i. Thermo flask filled with hot tea
   ii. Radiator of car
   iii. Fan

   c. The temperature scale of a certain thermometer is given by a relation $t = a \ln(X)+b$, where a and b are constants, and X is the thermometric property of the fluid in the thermometer. If at the ice point and steam point the thermometric property are found to be 1.5 and 7.5 respectively. What will be the temperature corresponding to the thermometric property 3.5. **07**

2. a. Starting from a convenient common state point on a P-V diagram, show the four expansion process for n=0, n=1, n= γ (where γ is the ratio of specific heats) and n=∞. What is each process called? Indicate their names adjacent to the processes diagram. **04**

   b. Write the similarities and dissimilarities between heat and work. **03**

   c. To a closed system 150KJ of work is supplied. If the initial volume is $0.6m^3$ and pressure of the system is changes as P= 8-4V, where V is the Volume in m^3 and pressure is in bar. Determine the final volume and the pressure of the system. **07**

3. a. Show that energy is a property of the system. Define the specific heats at constant volume and constant pressure **07**

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b. A piston and cylinder machine contains a fluid system, which passes through a complete cycle of four processes. During the cycle, the sum of all heat transfer is -170 kJ. The system completes 100 cycles per minute. Complete the following table showing the method for each item, and compute the net rate of work output in kW. **07**

Process	Q(kJ/min)	W(kJ/min)	ΔE(kJ/min)
a-b	0	2170	-
b-c	21000	0	-
c-d	-2100	-	-36000
d-a	-	-	-

4. a. Write the steady flow energy equation and explain all the terms involved. And simplify it for the following cases. **06**
i. insulated turbine, ii. throttle valve, iii. adiabatic nozzle

b. Air flows steadily at the rate of 0.5kg/s through an air compressor entering at 7m/s, with a pressure of 100kPa and a specific volume of $0.95m^3/kg$ and leaving at 5m/s, with a pressure of 700kPa and specific volume of $0.19m^3/kg$. The internal energy of the air leaving of 93kJ/kg greater than that of air entering. Cooling water in the compressor jackets absorbs heat from the air at a rate of 58kW. Compute the rate of shaft work input to the air in kW and find the ratio of input pipe diameter to the output pipe diameter. **08**

5. a. Prove that $COP_{HP}=COP_R+1$. **03**

b. State the Clausius and Kelvin- Planck's statements of the second law. **04**

c. A reversible heat engine operates between two reservoirs at temperatures of 600^0C and 40^0C. The engine drives a reversible refrigerator, which operates between 40^0C and -20^0C. The heat transfer to the engine is 2000kJ and network output from combined engine and refrigerator system is 360kJ. Calculate heat transfer to the refrigerator and net heat transfer to the reservoir at 40^0C. **07**

6. a. State and prove the Clausius inequality. **07**

b. Air at 20^0C and 1.05bar occupies $0.025m^3$. The air is heated at constant volume until the pressure is 4.5bar, and then cooled at constant pressure back to original temperature. Calculate: i) the net heat flow from the air ii) the net entropy change. Sketch the process in T-S diagram. Assume $C_p=1.005kJ/kgK$, $C_v= 0.718kJ/kgK$. **07**

7. a. A pressure cooker contains 1.5kg of steam at 5bar and 0.9 dryness fraction when the gas was switched off. Determine the quantity of heat rejected by the pressure cooker when the pressure in the cooker falls to 1bar. **08**

b. With a neat sketch, explain the method of measurement of dryness fraction of steam, using a throttling calorimeter. **06**

8. a. Explain the following: **06**

i. Compressibility factor

ii. Law of corresponding states

iii. Generalized compressibility chart

b. A fluid undergoes a reversible adiabatic compression from 0.5MPa, $0.2m^3$ to $0.05m^3$ according to the law $PV^{1.3}$=constant. Determine the change in enthalpy, internal energy and entropy and the heat transfer and work transfer during the process. **08**